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Importance of strength training to prevent secondary lower limb musculoskeletal injury in military personnel

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ABSTRACT

Considerable operational and financial costs are associated with musculoskeletal injury (MSKI) across military organisations. Following an MSKI, reaching 'medically fully deployable' status is often contingent on meeting physical employment standards. Strength training has a critical role to play in military physical preparation, injury mitigation and guiding rehabilitation practice. Low levels of muscle strength are associated with increased MSKI risk, and previous MSKI is a leading risk factor for re-injury. When an MSKI does occur, it is essential that all attempts to mitigate the risk of secondary MSKI are met. Maximal force (muscle strength) and rapid force (rate of force development and reactive strength) production have been identified as important strength characteristics requiring targeted training interventions and assessment prior to returning to operational roles. This review investigates the importance of strength training and its assessment during late-stage rehabilitation to inform return-to-duty exit criteria, meet physical employment standards and mitigate secondary MSKI.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Muscle strength is a vital determinant of military performance directly related to optimising physical preparation, injury mitigation strategies and guiding rehabilitation practice.

⇒ Low muscle strength and previous musculoskeletal injury (MSKI) are among the primary risk factors for developing an MSKI.

WHAT THIS STUDY ADDS

⇒ The assessment of rapid force production (muscle strength), assessed via rate of force development and reactive strength, is currently not accessible to defence rehabilitation practitioners.

⇒ If deficits in muscle strength are not clearly identifiable based on return-to-duty exit criteria and post-rehabilitation monitoring, the risk of secondary MSKI will be increased.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The integration of force assessment technology alongside the development of best practice guidelines for late-stage rehabilitation in the UK Defence is warranted.

INTRODUCTION

Musculoskeletal injuries (MSKIs) are a leading cause of attrition, contributing to a high proportion of missed training/duty days, medical downgrading and discharges of military personnel.[1] Lower-limb injuries account for the largest proportion of reported MSKIs,[2] imposing substantial operational and financial costs. Importantly, from a leadership, strategic, operational and tactical command perspective, injury-related work absenteeism may lead to deferred missions or requests for external support, placing additional strain on the broader force structure. Due to lower-limb MSKI eliciting such a toll on operational readiness, military leaders across the UK Defence Medical Services consider 'rehabilitation following MSKI' and ensuring personnel are 'ready and able' to meet the demands of the British Armed Forces a strategic research priority.[3]

The goal of rehabilitation is to safely return personnel to their previous level of function and meet Service-specific physical employment standards (PES) and occupational requirements. An inability to pass PES due to inadequate levels of fitness will often result in reclassification of deployability status and future career limitations. Within military rehabilitation settings, tasks (or modified variations) from the PES should be incorporated both as functional exercises and as formal assessments to aid return-to-duty decision-making criteria because reaching 'medically fully deployable' status is often contingent on meeting these physical performance benchmarks (eg, the British Army soldier conditioning assessment).[4] Strength training has a critical role to play in military physical performance and meeting PES following MSKI.[4 5] Low levels of muscle strength are associated with increased MSKI risk[6] and a previous MSKI is a leading risk factor for re-injury.[7] This review will investigate the importance of strength training and its assessment during late-stage rehabilitation to inform return-to-duty criteria and mitigate secondary MSKI.

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ACUTE AND POST-ACUTE MSKI CHALLENGES

Muscle atrophy and weakness

Disuse of skeletal muscle, often associated with MSKI, can lead to rapid and progressive atrophy; shortening of muscle fibres, reduced muscle compliance and decreased oxidative capacity.[8] This decline particularly affects personnel requiring longer periods of immobilisation or reduced activity, such as bone fractures or postoperative recovery. In the acute phase following an injury, swelling, inflammation and joint laxity in peripheral tissue may change the discharge of sensory receptors, which causes neuromuscular inhibition.[9] Muscle inhibition may lead to persistent force production deficits such as reduced ballistic, maximal and rate of force development (RFD, the ability to rapidly increase force at the onset of muscle contraction) strength characteristics that impair function, delay return to operational performance and elevate risk of re-injury.[10] Early strategies to limit atrophy and elicit muscle hypertrophy and strength are therefore essential.

Tendon health

The primary role of the muscle-tendon unit is to act as one entity to store, recoil and release energy to transmit force. In the presence of Achilles tendinopathy, RFD, leg stiffness (the resistance to the deformation of the lower limb in response to an applied force) and modulations of the stretch-shortening cycle (SSC, the ability to change quickly from an eccentric to concentric muscle action) are adversely altered.[11] Restoration of full tendon health and performance relies on progressive mechanical loading building from 'low and slow' (ie, strength endurance) through to 'high and fast' (ie, plyometric training) (see table 1), with microcycles and pain-titrated progression.[12]

IMPORTANCE OF MUSCLE STRENGTH TO PREVENT SECONDARY LOWER-LIMB MSKI

Even when military personnel receive regular evidence-based physical therapy, military personnel with previous MSKIs can exhibit more residual muscle strength deficits than those without a history of MSKI.[13] Even when athletes follow standardised rehabilitation programmes, peak force is often returned to pre-injury status and they might be considered nearly 'fully recovered' based on subjective outcome measures, but RFD can still be reduced/impaired.[14] These persistent strength deficits highlight that these strength characteristics have either not been objectively assessed and targeted throughout rehabilitation and/or they are not monitored and managed routinely in the medium to long term following discharge from rehabilitation care. The principles of strength and conditioning towards exercise programme design/delivery and neuromuscular performance testing are highly relevant and applicable when physically preparing personnel to return-to-duty.[4]

Maestroni and colleagues[9] outlined the underpinning principles of returning athletes to high-performance sport and the importance of strength and power training in MSK rehabilitation. Despite the population being athletes, their findings are

Table 1 Characteristics of different muscle strength qualities

Strength quality	Definition	Typical RM/intensity range	Example exercises	Common assessment methods	Military relevance
Maximum strength	The highest amount of force a muscle or muscle group can generate during a single voluntary contraction, regardless of the time taken to produce it. Typically assessed via 1RM testing in multi-joint lifts	~1–5 (≥85%)	Deadlift, back squat, lunges	1RM testing: *Dynamic tasks:* deadlift, back squat, bench press. *Isometric tasks:* isometric midthigh pull. HHD for isolated joints	Heavy casualty drags/carries. Load carriage with full kit. Obstacle climbing under load
Strength-speed	The ability to produce high levels of force at relatively fast velocities, usually against moderate to heavy loads moved explosively. The force-dominant side of ballistic and weightlifting movements	~3–6 (~70%–85%, moving explosively)	Weightlifting exercises and their derivatives (eg, clean, jerk and snatch variations), jump squats with moderate to high load	Weightlifting exercises (eg, power clean) (1RM or sub-max loads). Loaded jump squat with moderate loads (~50% 1RM)	Lifting/moving equipment quickly. Explosive starts with gear. Dynamic breaching tasks
Peak power	The point on the force–velocity curve where the product of force×velocity is maximised. The optimal mix of strength and speed for producing the highest mechanical power output	~30%–80% (load that maximises force×velocity)	Loaded jump squats, weightlifting derivatives (power clean, hang power snatch)	Loaded jump squat across different loads (force plate/bar velocity). Power clean (peak Watts calculated based on barbell velocity using a linear position transducer). Wingate anaerobic power test (cycling)	Short sprints with load. Obstacle vaults/jumps. Grenade/equipment throws
Speed-strength	The ability to generate force rapidly against light external loads at high movement velocities. The velocity-dominant side of explosive actions, often developed through plyometrics and light-load ballistic training	0%–40% (very light load, high velocity)	Plyometric jumps, bounding, depth jumps, light sled sprints	Jump/plyometric tests: Countermovement jump, squat jump, drop jump, hopping	Fire-and-move sprinting. Close combat explosive actions. Rapid changes of direction
Maximum speed	The highest movement velocity attainable, typically expressed as maximal sprinting speed. Relies heavily on neuromuscular efficiency, elastic energy utilisation and technical efficiency, with minimal external loading	Bodyweight only (no load)	Flying sprints, assisted sprints, sprint drills, overspeed training	Sprint tests (30–60 m, laser timing). Flying sprint (10–20 m split at max speed). GPS tracking of top speed	Pursuit/evasion sprinting. Fast battlefield repositioning. Time-critical message/medic runs

HHD, hand-held dynamometry; RM, repetition maximum.

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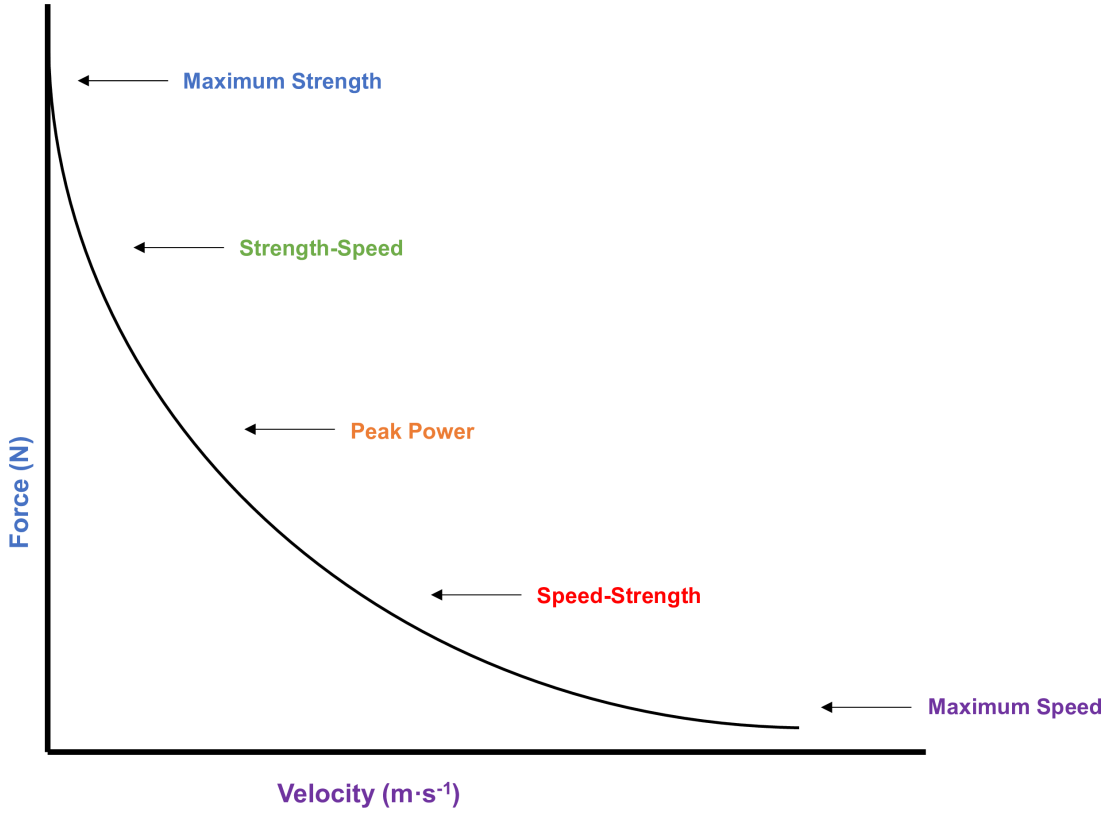


Figure 1 Force–velocity curve.

pertinent to military rehabilitation and physical preparation strategies (ie, mitigating secondary MSKI). Three key recommendations from this review[9] include: (1) targeting residual deficits in maximal strength, RFD and reactive strength (ie, SSC function) following injury to reduce the risk of future MSKI, (2) rehabilitation should prepare athletic populations to tolerate loads and velocities across the full spectrum of the force–velocity curve (see figure 1) and (3) through the completion of a comprehensive needs analysis, return-to-play (return-to-duty) criteria should consider multiple assessments of maximal strength, RFD and reactive strength capabilities along the whole force–velocity curve as dictated by the physical demands of a particular sport (military performance tasks) (see table 1). Considering these statements, the next section will explore the role of maximal force production (maximum strength) and rapid force production (RFD and reactive strength) in a military context and why these components are important when mitigating secondary MSKI risk.

Maximal force production

Maximal strength serves as the foundation of a soldier's ability to perform explosive movements necessary during close-quarter combat, jumping/landing, multidirectional speed, sprinting and throwing.[4] Thus, making the adoption of training with heavy loads advantageous, not only for relatively weaker personnel, but also for improving physiological characteristics necessary for high-velocity actions. Strength training with heavy loads (ie, ≥80% one repetition maximum) increases neural drive, intermuscular coordination, myofibrillar cross-sectional area of type II fibres, lean muscle mass and muscle fibre pennation angle.[15] Current evidence suggests that until individuals can squat ~1.6×body mass, maximal strength training should remain a dominant training modality in athletic populations.[16] It is not known how many military personnel reach this strength standard either when healthy or during later stages of rehabilitation. However, anecdotal evidence suggests maximal strength training should be considered a priority for many service personnel in physically demanding roles.

Despite its transfer to many functional tasks, maximal strength training largely consists of high forces at low velocities (see table 1). However, this alone may not fully prepare the musculoskeletal system to produce moderate-to-high forces rapidly, which underpin many military tasks. While there is strong interplay and overlap in both physiological determinants and performance between maximal strength training and ballistic training, they both induce different physiological adaptations.[9] Once appropriate levels of maximum strength have been attained following MSKI, there should be a greater training emphasis towards the development of rapid force production (via RFD and reactive strength training).

Rapid force development

Ballistic training increases RFD, movement velocity, jump height and sprint performance via lowered motor unit recruitment thresholds, improved motor unit firing frequency and synchronisation, as well as enhanced intermuscular coordination.[16] RFD is important as it is the force relative to the mass being moved that determines acceleration, with the duration over which the force is applied that determines the resulting velocity. Considering that force application during military skills such as sprinting and jumping lasts approximately 30–200 ms,[17] RFD is a critical performance characteristic central to success in many military tasks (see table 1). If rapid force production training considerations, such as speed-strength (the ability to produce force rapidly against light-to-moderate loads—see table 1), were neglected during the later stages of rehabilitation, a seemingly

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'fully recovered' soldier might return-to-duty only to find they cannot keep up with the quick pace of operations or worse, suffer re-injury when forced into a sudden sprint, jump or landing. In weaker individuals, RFD may be better enhanced via higher load strength training,[18] further highlighting that strength development should be the initial emphasis in training and rehabilitation.

Reactive strength

Reactive strength is the capacity to rapidly produce force following rapid eccentric loading, reflecting the efficiency of the SSC, identified during rebound tasks such as a drop jump or hopping. An individual's reactive strength capacity is underpinned by relative maximal eccentric strength[19]; this reinforces the need for adequate levels of muscle strength prior to developing SSC capabilities in military populations. The factors that underpin an efficient SSC include the SSC pre-stretch, antagonistic co-contraction, utilisation of intrinsic muscle-tendon stiffness and involuntary reflex muscle activity.[20] Ballistic and plyometric-based training can enhance RFD, leg stiffness and the modulation of the SSC[21] and are therefore considered valuable tools during late-stage rehabilitation and reducing the risk of secondary MSKIs throughout a service career (see table 1).

A soldier who has restored multidirectional speed, sprint and jump performance is more likely to be retained in an operational role (particularly those in ground close combat roles) than one who can demonstrate 'moderate' levels of maximum strength and is pain-free at rest. Thus, addressing deficits in both maximal and rapid force production characteristics following MSKI is essential to mitigate secondary MSKI and optimise the number of military personnel fit for operational duty.[4 5]

ADJUNCT EFFECTS OF STRENGTH TRAINING ON CARDIORESPIRATORY FITNESS

Following MSKI or surgery, cardiorespiratory fitness can also be compromised due to reduced mobility, physical activity restriction and general deconditioning. To mitigate the effect of general or localised deconditioning following MSKI, the development and maintenance of cardiorespiratory fitness during rehabilitation is vital. Low cardiorespiratory fitness is considered a strong predictor of MSKI,[22] and even when the MSKI is resolved, cardiorespiratory fitness may be compromised,[23] impacting return-to-duty. High levels of cardiorespiratory fitness enable personnel to complete physically demanding tasks over a prolonged duration at lower relative intensity. Alongside muscle strength, this makes cardiorespiratory fitness assessments a cornerstone of military physical preparation/combat readiness strategies, injury mitigation strategies, long-term health promotion initiatives and guiding return-to-duty criteria.

Role of muscle strength in running endurance performance

Running (with and without load) over short and long distances is an essential military task. Running performance is considered vital to the effectiveness of the 'warfighter' and is regularly viewed as a barrier for reaching 'medically fully deployable' status post-MSKI. The classic model of endurance running suggests that performance (running speed) can be predicted based on three physiological pillars: (1) maximal oxygen uptake ($\dot{V}O_2$ max), (2) lactate threshold (which together determine the fraction of sustainable $\dot{V}O_2$ max for a given distance) and (3) running economy[24] which is enhanced via strength training.[25] More recently, a fourth physiological pillar of endurance running has been proposed that reflects the temporal changes in these three parameters that result from fatigue during prolonged exercise (a common feature of military training). This has been termed 'durability' or 'resilience'.[26]

A high running economy is important to military personnel because more economical runners can run at faster speeds for the same oxygen cost. The proposed mechanisms for improving running economy via strength training include an increase in maximal force production, reducing relative neuromuscular activation/recruitment and relative muscle force requirements at a given speed and tendon stiffness.[27] Lower extremity stiffness is important as it increases with running velocity. Recently, strength training has been shown to positively influence running economy 'durability' and increased high intensity time to exhaustion at the end of a 90-minute run in the heavy-intensity domain (ie, in a fatigued state) in well-trained male distance runners.[28] It has been theorised that adding strength training could induce a delay to neuromuscular fatigue and provide a glycogen-sparing effect, due to a later activation of type II fibres during prolonged running.[28] Complementing endurance running with strength training should therefore be considered 'normal/routine' practice as part of any military physical preparation and injury mitigation strategy.

HOW DO WE ENSURE REHABILITATION INTERVENTIONS HAVE BEEN SUCCESSFUL?

To reach 'medically fully deployable' status and prevent secondary lower-limb MSKI, this review has highlighted the importance of passing PES and tailoring exercise to meet any deficits in maximal strength, RFD and reactive strength characteristics. How PES and these strength qualities are integrated into rehabilitation is therefore very important. Table 1 provides a summary of different strength qualities, example exercises, methods of assessment and the relevance of these strength qualities to specific military task performance.

PES: informing return-to-duty criteria

PES represent the 'minimum' standards expected by a single Service branch to safely fulfil the job role and are age and sex neutral. PES are often task-based and typically broken down into components that require repeated loaded sprints (section attacks), long-duration loaded marches and manual handling, such as lifting, carrying, pushing/pulling, either in isolation or performed repetitively or for sustained periods. Thus, requiring multiple elements of fitness to pass. Despite the stated importance of PES being integrated into military rehabilitation practice, a recent UK Defence rehabilitation practitioner survey, Cassidy *et al*[23] indicated inconsistent PES test application and no consensus on the most relevant late-stage assessment battery, signalling variability in clinical practice. Also, strength standards derived from PES in recruits/trainees, such as the isometric mid-thigh pull test, are too low to inform rehabilitation return-to-duty criteria.[29] This reinforces the need for military organisations to collect healthy normative reference values of physical performance (based on age, sex and job roles) to directly inform physical preparation, injury mitigation strategies and guide rehabilitation practice.[5] Such an approach would be preferable compared with reaching a minimum PES and being issued a 'pass' or 'fail' score, which is not conducive to delivering individualised care or personalised feedback.

Despite many nations' PES being revised/updated within the past decade, MSKI occurrence among military personnel remains high. Thus, reinforcing the view that PES more closely represent a 'minimum' occupational 'health and

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Box 1 Future research aligned with strength training and testing to inform secondary MSKI prevention strategies

Exercise rehabilitation protocols

⇒ Longitudinal studies: there is a lack of long-term studies tracking injured personnel from the point of injury, through rehabilitation to fully deployable status and beyond. Wider use of follow-up data will enable research groups to comment on return-to-duty rates, reinjury incidence and long-term functional outcomes, increasing our understanding of secondary MSKI prevention.

⇒ Randomised controlled trials (RCTs) on exercise-based rehabilitation strategies: more RCTs are needed to compare the efficacy of different rehabilitation strategies/protocols, including: (1) strength versus endurance-focused programmes, (2) traditional versus virtual/tele-rehabilitation models, (3) tech-enabled rehabilitation versus conventional care, (4) maximum strength versus ballistic training.

⇒ Strength training programmes that optimise the 'grey area' of transitioning from clinical rehabilitation to mainstream PT and completing full operational duties. Only select military units have access to reconditioning PT—does access to the reconditioning PT pathway shorten the duration of time to reach medically fully deployable status?

⇒ Late-stage rehabilitation/operational performance: the end goal of rehabilitation is not the same for all personnel. Rehabilitation protocols could better address the unique operational demands of military service, for example, the differences between individuals in ground close combat roles (infantry soldiers who must be able to perform load carriage in austere environments) versus the requirements associated with some non-ground close combat roles who are in more sedentary employment (eg, cyber, intelligence and office-based roles).

⇒ Better define/benchmark 'role-specific' fitness standards. Profiling personnel against these standards will optimise rehabilitation practices and mitigate secondary MSKI.

⇒ Operational demands often limit the opportunity for service personnel to complete their rehabilitation programme in its entirety. Assessing the feasibility and effectiveness of mobile physical training equipment to facilitate ongoing rehabilitation when deployed away from the local unit is warranted.

Technology-enabled rehabilitation and physical preparation

There is currently an underutilisation of technology to inform rehabilitation practices and monitor physical preparation and injury mitigation strategies. Future research could investigate the following areas:

⇒ Wearable and digital health integration into strength training programmes—pilot and longitudinal studies are needed to validate the use of wearable sensors, mobile applications and remote monitoring platforms in tracking real-time MSKI recovery and preventing secondary MSKI and improving retention prospects.

⇒ Wider use of portable technology that enables the closing of the sophistication gap between lab and field-based assessments. For example, force assessment technologies (force platforms and strain gauges) and velocity-based training devices.

Continued

Box 1 Continued

Standardisation of reporting practices

Despite similar goals, the diversity in testing methods within and across military nations creates inconsistencies in assessment, interpretation and return-to-duty criteria. These discrepancies can hinder the establishment of universal fitness thresholds and create challenges for multinational deployments or interoperability assessments. Future research should investigate:

⇒ Validation of outcome measures aligned to return-to-duty criteria based on occupation-specific tasks.

⇒ Develop consensus within and between nations on how to best measure muscle strength (tests and metrics used) and how best to interpret this data based on different military-specific scenarios (eg, different phases of MSK rehabilitation, injury mitigation strategies, fatigue monitoring, benchmarking/profiling performance and future physical preparation strategies prior to operational deployments).

⇒ Standardising the methodological reporting practices associated with the growing use of portable force plate technology across military nations.[30]

⇒ The precise numbers of personnel affected by secondary MSKI are currently unclear. The development of a method for recording secondary MSKI data in the electronic health records would allow greater transparency and a method for defence statistics to better examine secondary MSKI occurrence.

Education

⇒ Adherence and compliance: An exercise intervention is only as effective as the patients' degree of compliance with a recommended programme. A common observation made within residential rehabilitation research is that while adherence is high when patients are under supervised care at a rehab unit, it is generally low when self-managing during 'free-living' or back at their local unit. Low adherence to exercise interventions is likely to impact negatively on rehab outcomes, delay return-to-duty duration, influence long-term physical health and in some cases create career retention concerns.

MSKI, musculoskeletal injury; PT, physical training.

fitness' standard as opposed to an 'occupational performance' standard. The latter being more pertinent to maximising physical preparation and injury mitigation strategies (including secondary MSKI).[5] This underlines the importance of facilitating higher levels of physical fitness compared with simply reaching/meeting minimum PES. By increasing the margin between a soldier's physical fitness and the demands of a military task (eg, passing PES and/or the British Army soldier conditioning assessment), higher levels of both muscle strength and cardiorespiratory fitness can reduce fatigue and the relative effort applied to everyday military activities, indirectly lowering the risk of secondary MSKI and promoting long-term retention of a service person's career.

Assessment of maximal strength, RFD and reactive strength

Standardised, valid and reliable rehabilitation outcome measures used to inform return-to-duty criteria are warranted.[4 5] A fundamental limitation of many field tests measuring maximal or rapid force production is that these methods restrict practitioners to a single outcome measure (eg, load lifted in kg, height jumped in metres). Although

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'outcome-based' metrics can be useful, they fail to provide more valuable information on movement strategies (jump phase duration during the countermovement jump, or RFD during an isometric midthigh pull or loading asymmetries), which may be more sensitive to training loads.[5] For example, whether the individual requires improvements in force or velocity capabilities (or both) to maximise their structured training methods.[4] Currently, UK Defence rehabilitation practitioners use clinical tests such as a triple hop and broad jump as an indicator of lower-limb rapid force production. However, these tests produce 'outcome-based' metrics only and are therefore inadequate at determining RFD or reactive strength deficits. Measuring force-time characteristics (via force plate technology) may facilitate a more comprehensive evaluation of neuromuscular performance and enable greater identification of strength deficits compared with existing assessment practices (ie, only recording the outcome-based metric (eg, jump height)).[5 30]

UK Defence rehabilitation practitioners currently have no capacity to accurately measure two of the three essential strength characteristics raised within this review (RFD and reactive strength). This highlights a substantial knowledge and capability gap in defence rehabilitation practice and for physical training staff, ensuring personnel are 'ready and able' to meet the demands of their occupational role. The authors of a recent review[30] found that while the use of force plates to measure force production characteristics is growing rapidly across military organisations, the choice of tests and metrics being used has been underexploited, especially during rehabilitation practice or long-term surveillance post-MSKI.

Importance of post-MSKI surveillance research

Post-injury surveillance research in military populations is critical for understanding recovery trajectories, optimising rehabilitation protocols and identifying personnel at risk of long-term disability or medical discharge. Implementing large-scale surveillance systems to track outcomes following MSKIs also enables the analysis of injury trends, intervention effectiveness and return-to-duty rates. The measurement of muscle force production characteristics within post-MSKI surveillance research is an important variable to capture alongside a wide range of other factors when informing MSK health following MSKI. Subsequently, by addressing current knowledge gaps and enhancing surveillance capabilities, military medicine can better support injured personnel and reduce preventable medical downgrading or discharge. Box 1 provides examples of future research aligned with strength training and testing to inform secondary MSKI prevention strategies.

CONCLUSIONS

Reducing secondary MSKI is critical for operational readiness and organisational sustainability. The development of physical fitness, particularly maximal and rapid force production characteristics, is a key modifiable factor influencing rehabilitation outcomes and secondary injury risk. Evidence supports integrating structured and progressive strength training into rehabilitation programmes to improve functional recovery and reduce medical discharge rates. The use of objective monitoring, which reflects the demands associated with occupational performance, may improve the quality of rehabilitation, accountability and appropriateness of discharge. We propose that if a greater emphasis were to be placed on rehabilitation practitioners facilitating the return of injured personnel to occupational 'performance' prior to personnel being considered MFD compared with a return to occupational 'health and fitness' (ie, meeting PES), risk of secondary MSKI could be reduced. Substantial knowledge gaps remain regarding optimal rehabilitation protocols and predictive fitness thresholds influencing recovery and secondary injury risk. Addressing these gaps through longitudinal studies (to assess functional outcomes, re-injury rates and career trajectories), technology integration and implementation research is necessary to prevent secondary injury and preserve operational capability.

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REFERENCES

1 Lovalekar M, Hauret K, Roy T, *et al*. Musculoskeletal injuries in military personnel-Descriptive epidemiology, risk factor identification, and prevention. *J Sci Med Sport* 2021;24:963–9.
2 Andersen KA, Grimshaw PN, Kelso RM, *et al*. Musculoskeletal Lower Limb Injury Risk in Army Populations. *Sports Med Open* 2016;2.
3 Woodgate P. Realising the ambition of the Defence Medical Services research strategy. *BMJ Mil Health* 2024;170:189–90.
4 Ladlow P, Conway D, Hayhurst D, *et al*. Integration of strength training into UK Defence Rehabilitation practice: current trends and future challenges. *BMJ Mil Health* 2022;168:314–9.
5 Ladlow P, Suffield C, Greeves JP, *et al*. How "STRONG" is the British Army? *BMJ Mil Health* 2024;170:283–6.
6 de la Motte SJ, Gribbin TC, Lisman P, *et al*. Systematic Review of the Association Between Physical Fitness and Musculoskeletal Injury Risk: Part 2-Muscular Endurance and Muscular Strength. *J Strength Cond Res* 2017;31:3218–34.

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7 Kucera KL, Marshall SW, Wolf SH, *et al*. Association of Injury History and Incident Injury in Cadet Basic Military Training. *Med Sci Sports Exerc* 2016;48:1053–61.
8 Slysz J, Stultz J, Burr JF. The efficacy of blood flow restricted exercise: A systematic review & meta-analysis. *J Sci Med Sport* 2016;19:669–75.
9 Maestroni L, Read P, Bishop C, *et al*. Strength and Power Training in Rehabilitation: Underpinning Principles and Practical Strategies to Return Athletes to High Performance. *Sports Med* 2020;50:239–52.
10 Pietrosimone BG, Lepley AS, Ericksen HM, *et al*. Neural Excitability Alterations After Anterior Cruciate Ligament Reconstruction. *J Athl Train* 2015;50:665–74.
11 Wang H-K, Lin K-H, Wu Y-K, *et al*. Evoked spinal reflexes and force development in elite athletes with middle-portion Achilles tendinopathy. *J Orthop Sports Phys Ther* 2011;41:785–94.
12 Judd A, Wild K, Puxley L, *et al*. UK Defence Rehabilitation consensus agreement for the conservative management of Achilles and patellar tendinopathy: a modified Delphi approach. *BMJ Mil Health* 2025.
13 Garcia RCF, Muniz AM de S, Jacinto D de C, *et al*. Prior Musculoskeletal Injury and Components of Physical Fitness in Military Personnel: A Systematic Review with Meta-analysis. *Mil Med* 2025;190:e978–86.
14 Angelozzi M, Madama M, Corsica C, *et al*. Rate of force development as an adjunctive outcome measure for return-to-sport decisions after anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther* 2012;42:772–80.
15 Suchomel TJ, Nimphius S, Stone MH. The Importance of Muscular Strength in Athletic Performance. *Sports Med* 2016;46:1419–49.
16 Cormie P, McGuigan MR, Newton RU. Adaptations in athletic performance after ballistic power versus strength training. *Med Sci Sports Exerc* 2010;42:1582–98.
17 Taber C, Bellon C, Abbott H, *et al*. Roles of Maximal Strength and Rate of Force Development in Maximizing Muscular Power. *Strength Cond J* 2016;38:71–8.
18 Comfort P, Jones PA, Thomas C, *et al*. Changes in Early and Maximal Isometric Force Production in Response to Moderate- and High-Load Strength and Power Training. *J Strength Cond Res* 2022;36:593–9.
19 Beattie K, Carson BP, Lyons M, *et al*. The Relationship Between Maximal Strength and Reactive Strength. *Int J Sports Physiol Perform* 2017;12:548–53.
20 Pedley JS, Lloyd RS, Read P, *et al*. Drop Jump: A Technical Model for Scientific Application. *Strength Cond J* 2017;39:36–44.
21 Haff GG, Nimphius S. Training Principles for Power. *Strength Cond J* 2012;34:2–12.
22 Lisman PJ, de la Motte SJ, Gribbin TC, *et al*. A Systematic Review of the Association Between Physical Fitness and Musculoskeletal Injury Risk: Part 1-Cardiorespiratory Endurance. *J Strength Cond Res* 2017;31:1744–57.
23 Cassidy RP, Masters N, Rodgers R, *et al*. Current perspectives of late-stage rehabilitation and return to duty criteria following lower limb musculoskeletal injury in military personnel: a practitioner survey. *BMJ Mil Health* 2025.
24 Joyner MJ. Modeling: optimal marathon performance on the basis of physiological factors. *J Appl Physiol (1985)* 1991;70:683–7.
25 Blagrove RC, Howatson G, Hayes PR. Effects of Strength Training on the Physiological Determinants of Middle- and Long-Distance Running Performance: A Systematic Review. *Sports Med* 2018;48:1117–49.
26 Jones AM. The fourth dimension: physiological resilience as an independent determinant of endurance exercise performance. *J Physiol* 2024;602:4113–28.
27 Fletcher JR, MacIntosh BR. Running Economy from a Muscle Energetics Perspective. *Front Physiol* 2017;8.
28 Zanini M, Folland JP, Wu H, *et al*. Strength Training Improves Running Economy Durability and Fatigued High-Intensity Performance in Well-Trained Male Runners: A Randomized Control Trial. *Med Sci Sports Exerc* 2025;57:1546–58.
29 Walters V, Coppack RJ, Cassidy RP, *et al*. Use of an isometric mid-thigh pull test during musculoskeletal rehabilitation: can the criterion values from the updated British Army physical employment standards be used to inform UK Defence Rehabilitation practice? *BMJ Mil Health* 2022;168:279–85.
30 Ladlow P, Lunt KM, Comfort P, *et al*. The Use of Force Plate Technology to Measure Force Production Characteristics in Military Personnel: A Scoping Review of Methodological Reporting Practices. *Sports Med Open* 2025;11.

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